FINBANK

Consumer Loan Portfolio Analysis

SQL Exploratory Data Analysis | January to September 2024

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Tools: SQL Server | 8 Analytical Queries | Portfolio: 25 Loans

NOTE This analysis covers 25 loan records across 5 branches and 10 agents from January to September 2024. August and September data appears incomplete - findings from those months should be interpreted with caution pending full data extraction.

Introduction

This document presents a full SQL-based exploratory data analysis of FinBank's consumer loan portfolio for the period January to September 2024. The analysis was conducted as part of a structured data analytics training project, designed to simulate a real-world business intelligence assignment.

A junior data analyst was tasked with answering eight key business questions on behalf of the portfolio management team. Each question was answered using structured SQL queries written in SQL Server Management Studio (SSMS), with results interpreted through a business lens and translated into actionable recommendations.

About the Dataset

The FinBank database contains five related tables covering loan applications, customer demographics, agent and branch records, and repayment history. The dataset includes 25 loan records across 20 unique customers, processed by 10 agents across 5 branches in Kenya.

Table	Key Fields	Description
Customers	customer_id, full_name, credit_score, employment_status	Demographics and credit profile of each borrower
Loans	loan_id, loan_type, loan_amount, status, application_date	Core loan records including product type, value, and current status
Repayments	repayment_id, loan_id, payment_status, amount_paid	Payment history with on time, late, and missed classifications
Agents	agent_id, agent_name, team, branch_id	Agent records linked to branches and teams
Branches	branch_id, branch_name, region	Five branches across Nairobi, Coast, Western, and Rift Valley

Tools Used

- SQL Server Management Studio (SSMS) - all queries written and executed in SQL Server
- Microsoft Word - final report documentation and presentation

Business Questions Investigated

- Query 1 - Portfolio Health Overview: loan distribution by status, count and value percentages
- Query 2 - Repayment Behaviour: on time, late, and missed payment rates across the portfolio
- Query 3 - Default Rate by Loan Type: which products carry the highest default risk
- Query 4 - Agent Performance: loans processed, value disbursed, and processing speed per agent
- Query 5 - Monthly Application Trend: how application volume has changed month by month
- Query 6 - Repeat Borrowers: customers with multiple loans and time between borrowing cycles
- Query 7 - Branch and Region Performance: loan value and default rates across all five branches
- Query 8 - Agent Efficiency Ranking: agents ranked by average application to disbursement time

How to Read This Document

Each section follows the same structure: the business question is stated first in plain English, followed by the SQL query written to answer it, the result set returned by the database, and finally the analytical insight - what the data means for the business and what action is recommended.

A summary table of all eight findings and recommendations is provided on the final page for quick reference.

Query 1 Portfolio Health Overview

Business Question

What is the overall health of the loan portfolio? How are loans distributed by status and what percentage does each represent by both count and value?

SQL Query

```
WITH LoansDisbursed AS (
  SELECT
    status,
    COUNT(*) AS TotalLoans,
    SUM(loan_amount) AS TotalLoanAmount
  FROM Loans
  GROUP BY status
)
SELECT
  status,
  TotalLoans,
  TotalLoanAmount,
  ROUND(TotalLoanAmount * 100.0 / SUM(TotalLoanAmount) OVER(), 2) AS StatusPercentage,
  ROUND(TotalLoans * 100.0 / SUM(TotalLoans) OVER(), 2) AS CountPercentage
FROM LoansDisbursed
ORDER BY StatusPercentage DESC;
```

Result Set

Status	Total Loans	Total Amount (KES)	Value %	Count %
Active	20	5,288,000	97.96%	80.00%
Defaulted	4	80,000	1.48%	16.00%
Closed	1	30,000	0.56%	4.00%

Insight

Active loans dominate by value (97.96%) but represent only 80% of count - meaning the active loans are significantly larger in size than concluded ones. Of the 5 loans that have run their full course (1 Closed + 4 Defaulted), 4 ended in default - an 80% default rate on concluded loans. The average defaulted loan is KES 20,000 versus KES 264,400 for active loans, confirming defaults are concentrated in small personal loans. Recommendation: introduce a minimum credit score of 600 and employment verification for all personal loan applications below KES 50,000.

Query 2 Repayment Behaviour

Business Question

How are customers repaying their loans? What percentage of repayments are On Time, Late, or Missed - and which customers are driving the problematic payments?

SQL Query

```
WITH RepaymentStatus AS (
  SELECT
    payment_status,
    COUNT(*) AS TotalPayments
  FROM Repayments
  GROUP BY payment_status
)
SELECT
  payment_status,
  TotalPayments,
  ROUND(TotalPayments * 100.0 / SUM(TotalPayments) OVER(), 2) AS RepaymentPercentage
FROM RepaymentStatus
ORDER BY RepaymentPercentage DESC;
```

Result Set

Payment Status	Total Payments	Percentage
On Time	27	67.50%
Late	7	17.50%
Missed	6	15.00%

Customer Breakdown - Repayment Detail

```
SELECT
  C.full_name,
  C.credit_score,
  C.employment_status,
  COUNT(CASE WHEN R.payment_status = 'On Time' THEN 1 END) AS OnTime,
  COUNT(CASE WHEN R.payment_status = 'Late'    THEN 1 END) AS Late,
  COUNT(CASE WHEN R.payment_status = 'Missed'  THEN 1 END) AS Missed
FROM Loans L
INNER JOIN Customers C ON L.customer_id = C.customer_id
INNER JOIN Repayments R ON L.loan_id = R.loan_id
GROUP BY C.full_name, C.credit_score, C.employment_status
ORDER BY Missed DESC;
```

Insight

32.5% of all repayments are either Late or Missed - more than triple the industry benchmark of 10%. All 4 customers with Missed payments share two characteristics: credit scores below 550 and unstable employment (3 Unemployed, 1 Self-employed). Critically, none of these customers made a single payment - indicating a credit approval failure, not a collections issue. Additionally, Diana Auma (credit score 560,

self-employed) has made one payment to date and it was late - she matches the early risk profile of confirmed defaulters and is recommended for proactive follow-up within 30 days.

Query 3 Default Rate by Loan Type

Business Question

Which loan products carry the highest default risk? For each loan type, what is the default rate as a percentage of loans issued?

SQL Query

```
SELECT
  loan_type,
  COUNT(*) AS LoansIssued,
  COUNT(CASE WHEN status = 'Defaulted' THEN 1 END) AS DefaultedLoans,
  ROUND(
    COUNT(CASE WHEN status = 'Defaulted' THEN 1 END) * 100.0 / COUNT(*),
  2) AS DefaultRatePct
FROM Loans
GROUP BY loan_type;
```

Result Set

Loan Type	Loans Issued	Defaulted	Default Rate
Business Loan	9	0	0.00%
Mortgage	4	0	0.00%
Personal Loan	12	4	**33.33%**

Insight

Personal loans carry a 33% default rate - 1 in every 3 personal loans issued is not being repaid. Business loans and mortgages show zero defaults. This is not a coincidence: mortgages and business loans are secured against collateral and issued to higher credit score customers, while personal loans are unsecured and have been approved for customers with credit scores as low as 460. The credit screening criteria for personal loans must be stricter than for secured products, not looser. Recommendation: introduce a minimum credit score of 600 and mandatory employment verification for all personal loan applications.

Query 4 Agent Performance

Business Question

How is each agent performing? Show total loans processed, total value disbursed, and average days from application to disbursement per agent.

SQL Query

```
WITH LoansProcessed AS (
  SELECT
    agent_id,
    COUNT(*) AS TotalLoansProcessed,
    SUM(loan_amount) AS TotalLoanAmount,
    AVG(DATEDIFF(DAY, application_date, disbursement_date)) AS AvgDays
  FROM Loans
  GROUP BY agent_id
)
SELECT
  A.agent_name,
  A.team,
  B.branch_name,
  LP.TotalLoanAmount,
  LP.TotalLoansProcessed,
  LP.AvgDays
FROM Agents A
INNER JOIN Branches B ON A.branch_id = B.branch_id
LEFT JOIN LoansProcessed LP ON A.agent_id = LP.agent_id
ORDER BY TotalLoanAmount DESC;
```

Result Set

Agent	Team	Branch	Value (KES)	Loans	Avg Days
Carol Njeri	A	Westlands	**1,840,000**	3	8
Brian Otieno	B	Nairobi CBD	1,505,000	3	6
Alice Mwangi	A	Nairobi CBD	1,185,000	**5**	4
Esther Wanjiku	A	Mombasa	348,000	3	5
Grace Akinyi	A	Kisumu	205,000	4	**3**
David Kimani	B	Westlands	-	-	-

Insight

Agent performance varies across three dimensions - speed, volume, and value. Carol Njeri leads on value (KES 1.84M) handling large mortgages and business loans. Alice Mwangi leads on volume (5 loans). Grace Akinyi leads on speed (3 days average). David Kimani in Westlands has processed zero loans - this requires immediate investigation as Westlands is a high-demand Nairobi business district. Carol's 8-day average is not a performance concern - it reflects the additional verification required for high-value secured products. Recommendation: define agent KPIs by role type rather than applying a single efficiency metric across all agents.

Query 5 Monthly Application Trend

Business Question

How has loan application volume trended month by month from January to September 2024? Is the business growing, declining, or flat?

SQL Query

```
SELECT
  FORMAT(application_date, 'yyyy-MM') AS ApplicationMonth,
  COUNT(*) AS TotalApplications
FROM Loans
GROUP BY FORMAT(application_date, 'yyyy-MM')
ORDER BY MIN(application_date);
```

Result Set

Month	Total Applications
2024-01	3
2024-02	4
2024-03	4
2024-04	4
2024-05	4
2024-06	2
2024-07	2
2024-08	1
2024-09	1

Insight

Applications held steady at 3 to 4 per month from January to May 2024 before declining sharply from June onwards. Two possible business explanations are: (1) seasonal demand cycles aligned with school fees and harvest seasons reducing loan demand in the second half of the year, and (2) interest rate sensitivity causing customers to seek alternatives such as SACCOs or mobile loans. However August and September each show only 1 application - this is almost certainly incomplete data rather than a genuine business trend. Full year data must be confirmed before drawing conclusions on the second half decline.

Query 6 Repeat Borrowers

Business Question

Which customers have taken more than one loan? What is the total value they have borrowed and how many days passed between their first and most recent loan?

SQL Query

```
WITH CustomersLoans AS (
  SELECT
    customer_id,
    COUNT(*) AS TotalLoans,
    SUM(loan_amount) AS TotalLoanAmount,
    MIN(application_date) AS FirstApplicationDate,
    MAX(application_date) AS RecentApplicationDate
  FROM Loans
  GROUP BY customer_id
  HAVING COUNT(*) > 1
)
SELECT
  C.full_name,
  CL.TotalLoanAmount,
  CL.TotalLoans,
  DATEDIFF(DAY, CL.FirstApplicationDate, CL.RecentApplicationDate) AS DaysBetweenLoans
FROM Customers C
INNER JOIN CustomersLoans CL ON C.customer_id = CL.customer_id
ORDER BY TotalLoanAmount DESC;
```

Result Set

Customer	Loans	Total Amount (KES)	Days Between
Daniel Njoroge	2	**1,450,000**	139
Susan Wairimu	2	1,090,000	150
Samuel Kariuki	2	230,000	155
Winnie Achieng	2	105,000	138
John Omondi	2	70,000	161

Insight

FinBank has 5 repeat borrowers out of 20 unique customers - a 25% repeat rate. The average gap between first and most recent loan is approximately 150 days. Repeat borrowers represent significantly higher lifetime value - Daniel Njoroge alone has borrowed KES 1,450,000 across two loans compared to the single-loan average of KES 211,520. Recommendation: implement a targeted re-engagement campaign triggered at 120 days post-disbursement, offering pre-approved credit to customers with a clean repayment record. The 75% of customers who have not returned represent the primary growth opportunity within the existing customer base.

Query 7 Branch & Region Performance

Business Question

How does each branch and region perform on total loans, total value, and default rate? Which branches are generating value and which represent credit risk?

SQL Query

```
SELECT
  B.branch_name,
  B.region,
  COUNT(L.loan_id) AS TotalLoans,
  COUNT(CASE WHEN status = 'Defaulted' THEN 1 END) AS DefaultedLoans,
  ROUND(COUNT(CASE WHEN status = 'Defaulted' THEN 1 END) * 100.0 / COUNT(L.loan_id), 2)
AS DefaultRatePct,
  SUM(L.loan_amount) AS TotalLoanAmount
FROM Loans L
INNER JOIN Agents A ON L.agent_id = A.agent_id
INNER JOIN Branches B ON A.branch_id = B.branch_id
GROUP BY B.branch_name, B.region;
```

Result Set

Branch	Region	Loans	Value (KES)	Default Rate
Nairobi CBD	Nairobi	**8**	**2,690,000**	0.00%
Westlands	Nairobi	3	1,840,000	0.00%
Mombasa	Coast	4	438,000	0.00%
Kisumu	Western	5	315,000	0.00%
Nakuru	Rift Valley	5	115,000	**80.00%**

Insight

Nakuru branch carries an 80% default rate - 4 out of 5 loans issued have defaulted, representing KES 80,000 in potentially unrecoverable value against KES 115,000 total disbursed. At this default rate Nakuru is likely a net loss branch when operational costs are included. All other branches show zero defaults. The pattern is consistent with earlier findings: Nakuru agents approved small personal loans to customers with credit scores below 550 and unstable employment. Nairobi CBD by contrast disbursed KES 2,690,000 with zero defaults and should serve as the credit quality benchmark. Recommendation: mandatory credit retraining for Nakuru agents and immediate suspension of personal loan approvals for customers with credit scores below 600 in the Rift Valley region.

Query 8 Agent Efficiency Ranking

Business Question

How do agents rank against each other on processing speed - from application date to disbursement date? Who is fastest and who needs improvement?

SQL Query

```
WITH AgentAvgProcessing AS (
  SELECT
    agent_id,
    AVG(DATEDIFF(DAY, application_date, disbursement_date)) AS AvgDays
  FROM Loans
  GROUP BY agent_id
)
SELECT
  A.agent_name,
  B.branch_name,
  A.team,
  AP.AvgDays,
  DENSE_RANK() OVER (ORDER BY AP.AvgDays) AS AgentRank
FROM Agents A
INNER JOIN Branches B ON A.branch_id = B.branch_id
INNER JOIN AgentAvgProcessing AP ON A.agent_id = AP.agent_id
ORDER BY AgentRank;
```

Result Set

Agent	Team	Branch	Avg Days	Rank
Grace Akinyi	A	Kisumu	**3**	**1 - Fastest**
Irene Chebet	A	Nakuru	4	2
Alice Mwangi	A	Nairobi CBD	4	2
Esther Wanjiku	A	Mombasa	5	3
Felix Ochieng	B	Mombasa	5	3
James Rotich	B	Nakuru	5	3
Henry Maina	B	Kisumu	5	3
Brian Otieno	B	Nairobi CBD	6	4
Carol Njeri	A	Westlands	8	5 - Context required

Note: David Kimani excluded - no loans processed. A NULL average would rank first with DENSE_RANK, producing a misleading result.

Insight

Team A outperforms Team B on processing speed - averaging 4.8 days versus 5.3 days. Grace Akinyi ranks first at 3 days average. Carol Njeri ranks last at 8 days - but this is not a performance concern as she handles the most complex high-value products (mortgages, large business loans) which require additional verification steps. DENSE_RANK was used over ROW_NUMBER to correctly reflect tied performance -

Irene Chebet and Alice Mwangi both average 4 days and correctly share rank 2. Applying ROW_NUMBER would imply one outperforms the other when they are identical. Recommendation: define agent KPIs by product type - speed metrics for personal and small business loan agents, quality and value metrics for mortgage and large loan agents.

Summary of Findings

#	Query	Key Finding	Recommendation
1	Portfolio Health	80% default rate on concluded loans	Credit score minimum of 600
2	Repayment Behaviour	32.5% late or missed - 3x industry benchmark	Employment verification required
3	Default by Loan Type	Personal loans: 33% default rate	Tighten personal loan eligibility
4	Agent Performance	David Kimani: zero loans processed	Investigate immediately
5	Monthly Trend	Aug-Sep data likely incomplete	Confirm full year data extraction
6	Repeat Borrowers	25% repeat rate - 150 day avg gap	Re-engage at 120 days post-loan
7	Branch Performance	Nakuru: 80% default rate	Suspend approvals - retrain agents
8	Agent Efficiency	Team A faster: 4.8 vs 5.3 days	Define KPIs by product type

End of Report - FinBank Consumer Loan Portfolio Analysis 2024